Xue-Qiang Fan, Ph.D.

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About Me

I am currently a Ph.D. Candidate major in School of Computer and Information, Information and Communication Engineering, Hefei University of Technology.

- **Interests:** My research interests include Artificial Intelligence, Machine Learning, and Data Mining. In particular, I am interested in Pattern Recognition, Polarization Vision, Computational Imaging, and Bioinformatics.
- **Address:** Danxia Road 485, Shushan District, Hefei, Anhui, China.

Education

Jun. 2022 – Present
Ph.D. Hefei University of Technology, Hefei, China
Major: *Information and Communication Engineering, School of Computer and Information.* ***Advisor: Prof. Zhong-Yi Guo***

Jul. 2019 – May. 2022
M.S. Zhejiang University of Technology, Hangzhou, China
Major: *Control Science and Engineering, College of Information Engineering.* ***Advisor: Prof. Jun Hu***.

Sep. 2015 - Jun. 2019
B.S. Henan University of Urban Construction, Pingdingshan, China
Major: *Automation, School of Electrical and Control Engineering. Courses: Automatic Control Theory, etc..*

Research Experience

Zhong-Yi Guo Research Group, Advanced Electromagnetic Function Laboratory (AEMFLab)

June. 2022 – Present
Interests: Pattern Recognition, Polarization Vision, Computational Imaging.

Gui-Jun Zhang Research Group, Intelligent Optimization and Bioinformatics Lab (IOBIO-Lab)

July. 2019 - May. 2022
Interests: Pattern Recognition, Bioinformatics.

Research Publications

(# Contribution equally; * Corresponding authors)

Journal Articles (Featured Publications)

1. **X. Q. Fan**, W. Y. Chen, B. Lin, P. Peng, and Z. Y. Guo*, "Improved polarization scattering imaging using local-global context polarization feature learning framework," *Opt. Lasers Eng.*, vol. 178, p. 108 194, 2024. DOI: 10.1016/j.optlaseng.2024.108194.
2. **X. Q. Fan**, B. Lin, K. Guo, B. Y. Liu, and Z. Y. Guo*, "TSMPN-PSI: High-performance polarization scattering imaging based on three-stage multi-pipeline networks," *Opt. Express*, vol. 31, pp. 38 097–38 113, 2024. DOI: 10.1364/OE.501269.
3. **X. Q. Fan**, B. Lin, and Z. Y. Guo*, "Infrared polarization-empowered full-time road detection via lightweight multi-pathway collaborative 2d/3d convolutional networks," *IEEE Trans. Intell. Transp. Syst. In Press*, 2024. DOI: 10.1109/TITS.2024.3383405.

4. B. Lin, **X. Q. Fan**, P. Peng, and Z. Y. Guo*, "Dynamic polarization fusion network (DPFN) for imaging in different scattering systems," *Opt. Express*, vol. 32, pp. 511–525, 2024. DOI: 10.1364/OE.507711.

5. **X. Q. Fan**, B. Lin, J. Hu, and Z. Y. Guo*, "I-DNAN6mA: Accurate identification of dna n6-methyladenine sites using the base-pairing map and deep learning," *J. Chem. Inf. Model.*, vol. 63, no. 3, pp. 1076–1086, 2023. DOI: 10.1021/acs.jcim.2c01465.

6. B. Lin, **X. Q. Fan**, and Z. Y. Guo*, "Self-attention module in multi-scale improved u-net (SAM-MIU-net) motivating high-performance polarization scattering imaging," *Opt. Express*, vol. 31, no. 2, pp. 3046–3058, 2023. DOI: 10.1364/OE.479636.

7. **X. Q. Fan**, J. Hu*, Y. X. Tang, D. J. Yu*, and G. J. Zhang*, "Predicting rna solvent accessibility from multi-scale context feature via multi-shot neural network," *Analytical Biochemistry*, vol. 654, p. 114 802, 2022. DOI: 10.1016/j.ab.2022.114802.

8. **X. Q. Fan**, J. Hu*, N. X. Jia, D. J. Yu*, and G. J. Zhang*, "Improved protein relative solvent accessibility prediction using deep multi-view feature learning framework," *Analytical Biochemistry*, vol. 631, p. 114 358, 2021. DOI: 10.1016/10.1016/j.ab.2021.114358.

Miscellaneous Experience

Awards and Achievements

2021 **The National Scholarship**, Awarded by Zhejiang University of Technology.

2020 **The National Scholarship**, Awarded by Zhejiang University of Technology.

Skills

Programming Languages	Python, Linux Shell Script, Matlab, Java, LaTeX, …
Misc.	Academic research, LaTeX typesetting, …
Hobbies	Running, Road Cycling, Folk Music, Movie, …

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教育背景

合肥工业大学 | 计算机与信息学院 | 信息与通信工程专业 | 学术型博士研究生 2022.06—2026.06（预计）

主要研究方向为**偏振信息处理，深度学习**，特别关注**偏振成像探测应用**研究。**导师：郭忠义教授**.

浙江工业大学 | 信息工程学院 | 控制科学与工程专业 | 学术型硕士研究生 2019.07—2022.05

主要研究方向为**生物信息学，模式识别**，特别关注**蛋白质、RNA 和 DNA 相关生物化学属性、功能识别研究**，例如 DNA 甲基化位点预测。**导师：胡俊教授**.

河南城建学院 | 电气与控制工程学院 | 自动化专业 | 工学学士 2015.09—2019.06

主要学习的专业课程有**自动控制理论、电路原理、计算机语言及程序设计、数字/模拟电子技术基础、电机拖动与运动控制、电力电子技术基础、过程控制**等。

研究经历

郭忠义教授课题组, **Advanced Electromagnetic Function Laboratory (AEMFLab)**, **合肥工业大学** | 博士研究生 2022.6—今

- **研究兴趣包括人工智能、机器学习和数据挖掘，尤其对模式识别、偏振视觉、计算成像、目标检测、三维重建和生物信息学感兴趣**

张贵军教授课题组, **Intelligent Optimization and Bioinformatics Lab (IOBIO-Lab)**, **浙江工业大学** | 硕士研究生 2019.7—2022.6

- **主要开展蛋白质、RNA 和 DNA 相关生物化学属性、功能识别研究**

发表文章 (部分)

(# 共同一作; * 通讯作者)

- **Xue-Qiang Fan (樊学强)**, Bing Lin, Zhong-Yi Guo*. Infrared Polarization-Empowered Full-Time Road Detection via Lightweight Multi-Pathway Collaborative 2D/3D Convolutional Networks. IEEE Trans. Intell. Transp. Syst., 2024. [DOI: 10.1109/TITS.2024.3383405] (JCR1 区, 中科院 1 区 TOP, IF=8.5)
- **Xue-Qiang Fan (樊学强)**, Wei-Yun Chen, Bing Lin, Peng Peng, Zhong-Yi Guo*. Improved Polarization Scattering Imaging Using Local-Global Context Polarization Feature Learning Framework. Opt. Lasers Eng., 2024;178:108194. [DOI: 10.1016/j.optlaseng.2024.108194] (JCR1 区, 中科院 2 区, IF=4.6)
- **Xue-Qiang Fan (樊学强)**, Bing Lin, Kai Guo, Bing-Yi Liu, Zhong-Yi Guo*. TSMPN-PSI: High-performance polarization scattering imaging based on three-stage multi-pipeline networks. Opt. Express 31, 38097-38113 (2023). [DOI: 10.1364/OE.501269] (JCR2 区, 中科院 2 区 TOP, IF=3.8)
- **Xue-Qiang Fan (樊学强)**, Bing Lin, Jun Hu*, Zhong-Yi Guo*. I-DNAN6mA: Accurate Identification of DNA N6-Methyladenine Sites Using the Base-Pairing Map and Deep Learning. J. Chem. Inf. Model. 2023, 63, 3, 1076–1086. [DOI: 10.1021/acs.jcim.2c01465] (JCR1 区, 中科院 2 区 TOP, IF=5.6)
- **Xue-Qiang Fan (樊学强)**, Jun Hu*, Yu-Xuan Tang, Dong-Jun Yu*, Gui-Jun Zhang*. Predicting RNA solvent accessibility from multi-scale context feature via multi-shot neural network. Analytical Biochemistry, 2022, 654: 114802. [DOI: 10.1016/j.ab.2022.114802] (JCR2 区, 中科院 3 区, IF=3.6)
- **Xue-Qiang Fan (樊学强)**, Jun Hu*, Ning-Xin Jia, Dong-Jun Yu*, Gui-Jun Zhang*. Improved protein relative solvent accessibility prediction using deep multi-view feature learning framework. Analytical Biochemistry, 2021, 631: 114358. [DOI: 10.1016/j.ab.2021.114358] (JCR2 区, 中科院 3 区, IF=3.6)
- Bing Lin, **Xue-Qiang Fan (樊学强)**, Peng Peng, Zhong-Yi Guo*. Dynamic polarization fusion network (DPFN) for imaging in different scattering systems. Opt. Express 32, 511-525 (2024). [DOI: 10.1364/OE.507711] (JCR2 区, 中科院 2 区 TOP, IF=3.8)
- Bing Lin, **Xue-Qiang Fan (樊学强)**, Zhong-Yi Guo*. Self-attention module in multi-scale improved U-net (SAM-MIU-net) motivating high-performance polarization scattering imaging. Optics Express, Vol. 31, Issue 2, pp. 3046-3058 (2023). [DOI: 10.1364/OE.479636] (JCR2 区, 中科院 2 区 TOP, IF=3.8)
- 王晓飞, **樊学强**, 李章维 *. 基于迁移学习和多视图特征融合提高 RNA 碱基相互作用预测. 计算机科学, 2023, 50 (3): 164-172. [DOI: 10.11896/jsjkx.211200186]

授权专利

- 胡俊，**樊学强**，郑琳琳，董世健，白岩松，张贵军. 一种基于循环网络的蛋白质与蛋白质相互作用预测方法，授权日期：2024-03-26, 中国，授权号：ZL202110086831.5.
- 胡俊，**樊学强**，白岩松，郑琳琳，张贵军. 一种基于多视角学习的蛋白质溶剂可及性预测方法，授权日期：2022-07-26, 中国，授权号：202110558859.
- 胡俊，**樊学强**，董世建，白岩松，张贵军. 一种基于迭代搜索策略的蛋白质溶剂可及性预测方法，授权日期：2021-12-17, 中国，授权号：ZL202011030157.0.
- 胡俊，郑琳琳，**樊学强**，白岩松，张贵军. 一种基于多序列联配信息的配体绑定残基预测方法，授权日期：2021-10-29, 中国，授权号：ZL201911225431.7.
- 胡俊，白岩松，**樊学强**，郑琳琳，张贵军. 一种基于深度卷积神经网络的配体绑定残基预测方法，授权日期：2021-10-29, 中国，授权号：ZL201911225424.7.

主持或参加科研项目（课题）

- 基于深度多视角特征学习框架的蛋白质溶剂可及性预测，2021 年大学生建龙基金 (运河杯) 项目，结题，第一主要参与人.
- 蛋白质关键位点的深度学习方法研究与系统实现，2021 年大学生建龙基金 (运河杯) 项目，结题，参与.
- 基于深度学习的蛋白质生物数据挖掘研究，编号：KYY-ZX-20190503，结题，参与.

获得奖项

- 国家奖学金，浙江工业大学，2021/2020。
- 2021 年大学生建龙基金 (运河杯) 学术科技作品竞赛，一等奖 (第一负责人)，三等奖

技术能力

- **语言**: 编程不受特定语言限制。常用 Python, Matlab；熟悉 C, Java.
- **工作流**: Linux, Shell, GitHub.
- **其他**: 热衷跑步，公路骑行，(Folk) 音乐，电影.